

^{212}Bi α decay (60.55 min)

Type	Author	History Citation	Literature Cutoff Date
Full Evaluation	M. J. Martin	NDS 108,1583 (2007)	1-Jun-2007

Parent: ^{212}Bi : $E=0.0$; $J^\pi=1^{(-)}$; $T_{1/2}=60.55$ min 6; $Q(\alpha)=6207.26$ 3; $\% \alpha$ decay=35.94 6

^{212}Bi - $\% \alpha$ decay: Weighted average of 36.00 3 (1965Wa09), 35.81 4 (1962Be09), and 35.96 6 (1960Sc07) based on $I_\alpha(^{212}\text{Bi})/[I_\alpha(^{212}\text{Bi})+I_\alpha(^{212}\text{Po})]$. Others: 1961Ba12, 1961Fe04, 1958Ev80, 1958Pr70, 1956Se17, 1953Ma26 and 1950To63.

This data set is updated from 1992Ar05. The $Q(\alpha)$ value has been replaced by the value from 2003Au03. The following are J^π changes: $J^\pi(\text{g.s.})$ from $5^{(+)}$ to 5^+ , $J^\pi(40)$ from $4^{(+)}$ to 4^+ , $J^\pi(328)$ from $5^{(+)}$ to 5^+ , $J^\pi(473)$ from $(3^+, 4^+, 5^+)$ to (4^+) , and $J^\pi(493)$ from $(3^+, 5^+)$ to $(3)^+$. The mult data have been reanalyzed using the theoretical values of 2005KiZT.

 ^{208}Tl Levels

$(\gamma)(\gamma)$: 1958De25.

$(\alpha)(\gamma)$: 1955Ni19, 1962Be09, 1967Be19.

$(\alpha)(\gamma)(\theta)$: 1) 6051 $\alpha - 40\gamma$ data of 1955We10, 1956Fi41, 1956Ho11, and 1968Do17 are consistent with the spin sequences 1-3-4, and 1-4-5. Not consistent with 1-5-5, 1-4-4, 1-5-4, nor 1-6-5. Not consistent with $J(^{212}\text{Bi})=0$. The evaluator assumes that the 39.9 γ is M1 ($\delta < 0.02$ based on ce data). 2) 5768 $\alpha - 288\gamma$ and 5768 $\alpha - 328\gamma$ data of 1966KiZZ are consistent with the spin sequences 1(L,L+2)J(D,Q)4 and 1(L,L+2)J(D,Q)5, respectively, for $J(328 \text{ level})=5$, $L(\alpha)=5$, $\delta(288\gamma)=+0.067$ 6, $\delta(328\gamma)=+0.07$ 5. Other: 1963Co28. 3) 5625 $\alpha - 432\gamma$ and 5625 $\alpha - 472\gamma$ data of 1966KiZZ are not consistent with $J(472 \text{ level})=3$ to 6, the maximum J-range allowed by ce data. Authors' analysis, which yields $J=4$, $\delta(432\gamma)=+1.7$ to 2.6, $\delta(472\gamma)=-14$ to -8 , $\delta(\alpha L=5/L=3)=+1$, appears to be in error. In particular, the experimental A_4 terms have the opposite sign from the theoretical values (for all values of $L=5$, $L=3$ α -mixing), whereas 1966KiZZ show the theoretical A_4 values changing signs with α mixing (the authors' term linear in $\delta(\alpha)$ appears to be ≈ 5 times too large). The lack of agreement in possible J-values between the $\text{Ag}(\theta)$ and ce data might be due to the inability to resolve the weak 5626 α from the 5607 α in the α gate and the weak 432 and 472 γ 's from 452 γ in the coincidence spectrum. 4) 5606 $\alpha - 452\gamma$ and 5606 $\alpha - 492\gamma$ data of 1966KiZZ are consistent with the spin sequences 1(L,L+2)J(D,Q)4 and 1(L,L+2)J(D,Q)5, respectively, for $J(492 \text{ level})=3$, $L(\alpha)=3$, $\delta(453\gamma)=-0.048$ 8 and for $J=5$, $L(\alpha)=5$, $\delta(453\gamma)=+0.123$ 4, $\delta(492\gamma)=-0.51$ 2, $+2.06$ 8. Data are not consistent with $J=4$ or 6. Note, however, that the 492 γ is very weak. (others: 1956Ko60, 1963Co28 for 452 γ) 5) 5486 $\alpha - 620\gamma$ and 5486 $\alpha - 576\gamma$ data of 1966KiZZ are suspect. The 5486 α was not seen in the authors' spectrum, so a wide gate was set at its expected position. And the existence of the 576 γ is in doubt. IT has not been reported in singles studies and, as pointed out by 1966KiZZ, most of its contribution to $\text{Ag}(\theta)$ could be due to chance coincidences from the strong 583 γ in ^{208}Pb . Data for the 620 γ are consistent with $J(620 \text{ level})=6$, $L(\alpha)=5$, $\delta(620\gamma) < -8$; > 47 ; not consistent with $J=3, 4$ or 5.

$\text{Ag}(\theta, \text{H}, \text{t})$: 1969Va21 studied the transient magnetic field (due to recoil) on ^{208}Tl nuclei in Fe.

E(level) [†]	J^π [‡]	$T_{1/2}$	Comments
0.0	5^+	3.053 min 4	$T_{1/2}$: $(\alpha)(\text{ce}(L) 39.85\gamma)(\text{t})$; measured recoil of atoms (1965Se02). Others(ps): ≈ 4 (1956Bu26), 2.6 10 (1958Si81).
39.857 4	4^+	6.5 ps 8	
328.04 4	5^+	> 0.1 ps	
473.4 4	(4^+)		
492.84 4	(3^+)		
621#	(6^+)		
759#			
803#			

[†] From E_γ , unless otherwise noted.

[‡] From ^{208}Tl Adopted Levels.

Calculated from $\Delta Q(\alpha)$.

^{212}Bi α decay (60.55 min) (continued) α radiations

(α) γ (θ): see ^{208}Tl levels from ^{212}Bi α decay (60.55 min).

Other: [1962Be09](#),

$E\alpha^\dagger$	E(level)	$I\alpha^\dagger\&$	$HF^\ddagger@$	Comments
5302 2	803	0.00011 1	≈ 20000	
5345	759	0.0010 1	≈ 3800	
5481	621	0.013	≈ 1500	
5607	492.84	1.13	≈ 70	
5626	473.4	0.157	≈ 630	
5768	328.04	1.70	≈ 280	$I\alpha$: other: 1.67 2 (1962Be09).
6050.78 [#] 3	39.857	69.91 [#] 15	≈ 130	
6089.88 [#] 3	0.0	27.12 [#] 14	≈ 480	

[†] From [1960Wa14](#), unless otherwise noted. $E\alpha$ corrected (+4 keV) for calibration change (see [1991Ry01](#)). $I\alpha$ renormalized by evaluator to $\Sigma I\alpha(\text{weak branches})=100-\Sigma I\alpha(6051+6090)$.

[‡] $r_0(^{208}\text{Tl})=1.50$ 2.

[#] From [1991Ry01](#).

[@] The radius parameter is poorly known, and alone contributes an uncertainty of $\Delta(HF)=+53\%$ and -35% . $HF=71+38-25$, for the 5607α , for example.

[&] For absolute intensity per 100 decays, multiply by 0.3594 6.

 $\gamma(^{208}\text{Tl})$

Absolute intensity measurements: $I_\gamma(288.20\gamma)=0.337$ 4 per 100 ^{212}Bi decays ([1984Ge07](#)) which gives $I_\gamma(288.20\gamma)=0.938$ 8 per 100 ^{212}Bi α decays. Other: [1983Sc13](#).

$E_\gamma^\dagger$	$I_\gamma^\ddagger b$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. ^a	δ	α^c	Comments
39.857 4	2.96 24	39.857	4 ⁺	0.0	5 ⁺	M1		23.2	$\alpha(L)=17.81$ 25; $\alpha(M)=4.16$ 6; $\alpha(N+..)=1.275$ 18 $\alpha(N)=1.052$ 15; $\alpha(O)=0.204$ 3; $\alpha(P)=0.0193$ 3 E_γ : from the $B\times\rho$ measurements of 1956Si23 , recalculated by the evaluator using revised fundamental constants (1987Co39) and electron binding energies (1967Be73). Mult.: L1:L2:L3:M1:M2:M3:N12:O12=1000 25:106 5:8.2 33:262 15:30 5:6.2 25: 76 4:10 4 (1963Se20). L1:L2:L3:M1:N:O=1000:92:8.0:231:57:13 (1957Zh05). $\alpha=22.6$ 5 from $\alpha\gamma$ -coin (1966Il01). From L3/L1, $\delta<0.012$. E_γ : from 1973SlZJ ; observed ce(L) and ce(M) lines in ^{212}Pb source and found to be converted in Tl. Not reported by other authors. 1957Zh05 report $E=144.94$ with $I(\text{ce(K)})=0.036$. The transition is not seen by 1962Be09 who report $I_\gamma<0.006$. These data give $\alpha(K)_{\text{exp}}>6$, consistent only with a large E0 component.
^x 124.1									
^x 144.94 ^d									

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^{212}Bi α decay (60.55 min) (continued) $\gamma(^{208}\text{Tl})$ (continued)

E_γ [†]	I_γ ^{‡b}	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. ^a	δ	α^c	Comments
288.20 4	0.938 8	328.04	5 ⁺	39.857	4 ⁺	M1+E2	+0.067 6	0.436	$\alpha(\text{K})=0.357$ 5; $\alpha(\text{L})=0.0605$ 9; $\alpha(\text{M})=0.01411$ 20; $\alpha(\text{N}+..)=0.00432$ 6 $\alpha(\text{N})=0.00356$ 5; $\alpha(\text{O})=0.000692$ 10; $\alpha(\text{P})=6.54 \times 10^{-5}$ 10 Mult.: from K/L=5.6 6 (1967Be19); Ag(θ) (1963Co28, 1966K1ZZ). δ : δ from Ag(θ) (1966K1ZZ).
328.03 4	0.349 ^{&} 17	328.04	5 ⁺	0.0	5 ⁺	M1		0.307	$\alpha(\text{K})=0.252$ 4; $\alpha(\text{L})=0.0425$ 6; $\alpha(\text{M})=0.00990$ 14; $\alpha(\text{N}+..)=0.00303$ 5 $\alpha(\text{N})=0.00250$ 4; $\alpha(\text{O})=0.000486$ 7; $\alpha(\text{P})=4.60 \times 10^{-5}$ 7 Mult.: $\alpha(\text{K})_{\text{exp}}=0.230$ 15 (1955Ni19), 0.27 2 (1967Be19), 0.31 3 from (K x ray) γ/γ (1966I101). L1/K=0.23 +7-6 (1963Da11). L/K=0.15 4 (1967Be19), 0.21 4 (1955Ni19). Other: $\alpha(\text{K})_{\text{exp}}=0.30$ (1957Vo22).
433.7 5	0.047 [#] 9	473.4	(4 ⁺)	39.857	4 ⁺	M1		0.1451	$\alpha(\text{K})=0.1191$ 17; $\alpha(\text{L})=0.0199$ 3; $\alpha(\text{M})=0.00464$ 7; $\alpha(\text{N}+..)=0.001421$ 21 $\alpha(\text{N})=0.001172$ 17; $\alpha(\text{O})=0.000228$ 4; $\alpha(\text{P})=2.16 \times 10^{-5}$ 3 Mult.: $\alpha(\text{K})_{\text{exp}}=0.43$ 12 (1967Be19), 0.25 9 (1955Ni19). Other: 0.12 (1957Vo22), 0.13 7 (1963Da11) using authors Ice(K)/Ice(K)(328 γ).
452.98 5	1.010 9	492.84	(3 ⁺)	39.857	4 ⁺	M1		0.1292	$\alpha(\text{K})=0.1061$ 15; $\alpha(\text{L})=0.01772$ 25; $\alpha(\text{M})=0.00413$ 6; $\alpha(\text{N}+..)=0.001264$ 18 $\alpha(\text{N})=0.001042$ 15; $\alpha(\text{O})=0.000203$ 3; $\alpha(\text{P})=1.92 \times 10^{-5}$ 3 Mult.: $\alpha(\text{K})_{\text{exp}}=0.086$ 22 (1978Av01), normalized to mult(583 γ)=E2 In ^{208}Tl β^- decay, 0.115 6 (1967Be19), 0.100 7 (1955Ni19). Other: 0.11 (1957Vo22).
473.0 7	0.14 [#] 1	473.4	(4 ⁺)	0.0	5 ⁺	M1+E2		0.07 5	$\alpha(\text{K})=0.06$ 4; $\alpha(\text{L})=0.011$ 5; $\alpha(\text{M})=0.0027$ 10; $\alpha(\text{N}+..)=0.0008$ 3 $\alpha(\text{N})=0.00069$ 24; $\alpha(\text{O})=0.00013$ 5; $\alpha(\text{P})=1.1 \times 10^{-5}$ 6 Mult.: $\alpha(\text{K})_{\text{exp}}=0.079$ 18 (1967Be19), 0.100 15 (1966I101). 0.053 15 (1955Ni19). Other: 0.12 (1957Vo22). The value of 1966I101 is from (K x ray) γ/γ for the unresolved 473+493 γ ; however, the 493 γ is questionable.
493.3 ^d 7	<0.01 [#]	492.84	(3 ⁺)	0.0	5 ⁺				Mult.: Ice(K)=0.009 (1957Vo22) not seen by 1967Be19 who report $I_\gamma < 0.01$. These data give $\alpha(\text{K})_{\text{exp}} > 0.9$, requiring mult $\geq$ M4 or an E0 component. The placement is not consistent with mult=E0.

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^{212}Bi α decay (60.55 min) (continued) $\gamma(^{208}\text{Tl})$ (continued)

E_γ [†]	I_γ ^{‡b}	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Comments
576 ^{@d}	<0.001	621	(6 ⁺)	39.857	4 ⁺	I_γ : $I(\gamma)/I(620\gamma)=0.22$ 10 (1966KIZZ). Transition masked by strong 583 γ in ^{208}Tl decay.
620 ^{@d}	<0.003	621	(6 ⁺)	0.0	5 ⁺	I_γ : not seen. The I_γ limit is from 1962Be09.

[†] From 1957Vo22, unless otherwise noted. The evaluator has recalculated E_γ from $B \times \rho(\text{ce(K)})$ using revised fundamental constants (1987Co39) and electron binding energies (1967Be73). Others: 1984Ge07, 1982Sa36, 1973Da38, 1967Be19.

[‡] From 1984Ge07, unless otherwise noted. Others: 1983Sc13, 1982Sa36, 1973Da38, 1967Be19, 1962Be09, 1960Em01.

From 1967Be19.

@ From 1966KIZZ.

& Corrected for contribution from 327.6-keV γ from ^{232}U decay.

^a The quoted $\alpha(\text{K})_{\text{exp}}$ values are from Ice(K) and the adopted I_γ values, normalized to $\alpha(\text{K})=0.357$ for the 288 γ , unless noted otherwise.

^b For absolute intensity per 100 decays, multiply by 0.3594 6.

^c Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with Frozen orbital approximation based on γ -ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

^d Placement of transition in the level scheme is uncertain.

^x γ ray not placed in level scheme.

^{212}Bi α decay (60.55 min)

- Legend
- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\text{max}}$
 - $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\text{max}}$
 - $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\text{max}}$
 - $---$ γ Decay (Uncertain)
 - $\bullet$ Coincidence

Decay Scheme

Intensities: $I_{(\gamma+ce)}$ per 100 parent decays

